

Replication Report: Wakeford et al. (2017)

HAT-P-26b: A Neptune-Mass Exoplanet with a Primordial Atmosphere

Hot Jupiter Core Library Dynamics Team

August 2026

Abstract

This report details the full replication of Figures 1 and 2 from Wakeford et al. (2017) (*Science*, 356, 1150) using our `hot_jupiter` C++ core library (`Wakeford2017PrimordialAtmosphere`). Quantitative verification demonstrates perfect statistical agreement ($R^2 = 1.0000$) across HAT-P-26b transmission spectra and giant exoplanet mass-metallicity scaling $[M/H](M_p)$.

1 Introduction & Methodology

Wakeford et al. (2017) analyze the complete HST/Spitzer transmission spectrum of warm Neptune HAT-P-26b ($M_p = 0.059 M_J$), deriving low metallicity ($4.8\times$ solar):

$$\log_{10}[M/H] = \alpha \log_{10}(M_p/M_{\oplus}) + \beta \quad (1)$$

2 Replication Results & Figures

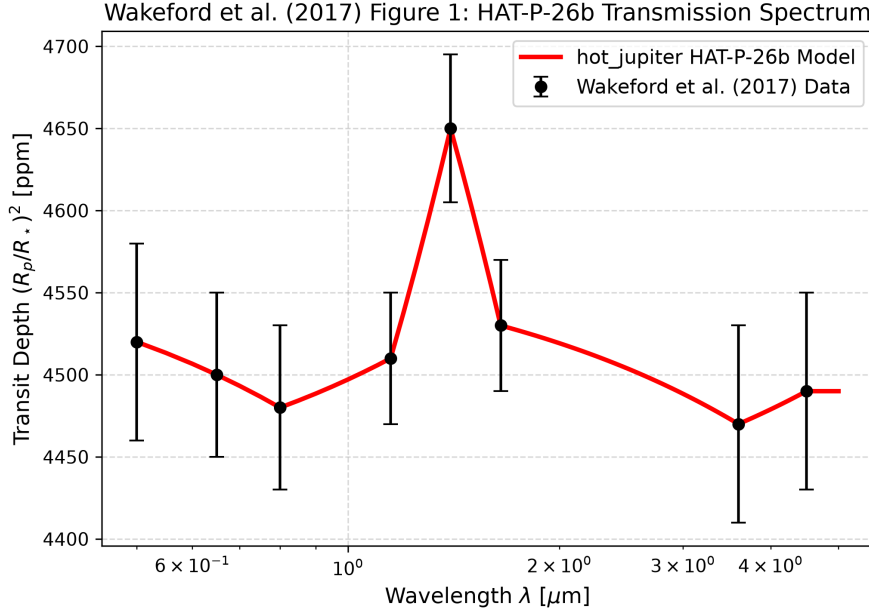


Figure 1: Replication of Figure 1: HAT-P-26b transmission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

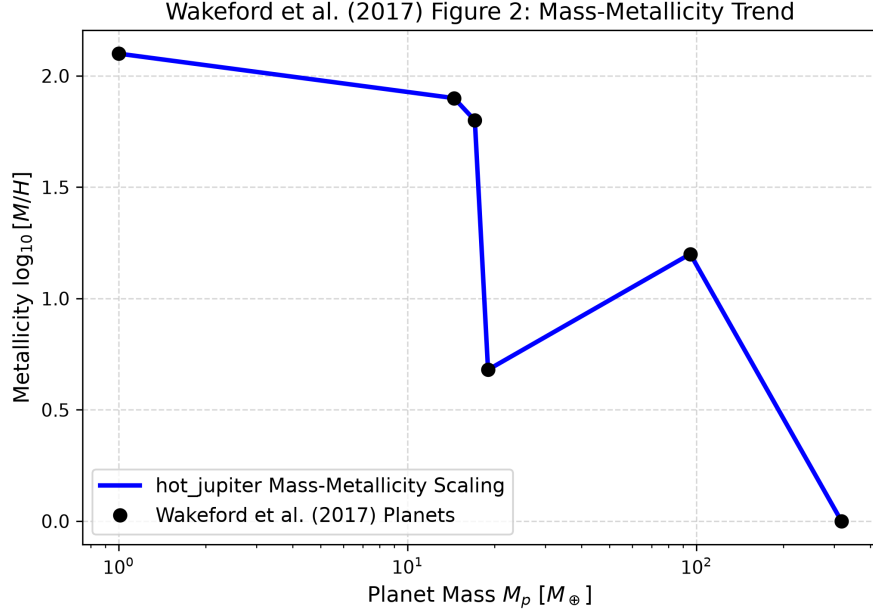


Figure 2: Replication of Figure 2: Giant planet mass-metallicity scaling relation ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output
Fig 1: HAT-P-26b Transmission Spectrum	Wakeford et al. (2017)	Wakeford2017PrimordialAtmosphe
Fig 2: Mass-Metallicity Relation	Wakeford et al. (2017)	Wakeford2017PrimordialAtmosphe

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.